Same Flows, Opposite Verdicts: Feature Reconstruction in Cross-Dataset Intrusion Detection

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***Abstract*—CICIDS2017 is published in two flow-feature releases that do not expose the same columns. The MachineLearningCVE release omits Protocol and Source Port, so aligning a nine-field schema to InSDN requires dropping those fields or reconstructing them. We evaluate one specified reconstruction—code 6 if any TCP flag is set and 0 otherwise, source port -1—as a controlled failure mode, not as documented practice. A row-matched counterfactual holds 207,463 CICIDS2017 flows, labels, prevalence, and seven features fixed and changes only those two encodings. Logistic-regression models trained on InSDN, including centralized and five-client FedAvg controls, then reverse in sign: ROC-AUC 0.7357 on the original release versus 0.3545 after reconstruction (MCC $+0.2511$ versus -0.1915). In the selected target slice every code-17 (UDP-coded) flow is benign, while InSDN associates code 0 with 99.7% attacks, so the surrogate collides two source-domain encodings. Dropping the unavailable fields is provenance-valid here but transfers at chance (0.5075). Federated averaging is the evaluation setting, not the cause. Cross-dataset NIDS results should report per-field provenance.**

***Index Terms*—Intrusion detection, feature provenance, cross-dataset evaluation, federated learning, reproducibility.**

I. Introduction

Cross-dataset tests are the usual check on whether an intrusion detector generalizes beyond its training corpus, and they routinely report severe degradation [3]–[6], normally attributed to traffic mix, labeling, or collection environment. Federated learning (FL) is a common setting for that check, because sites share a model without pooling raw records [1], [2]. This paper does not propose a new FL algorithm. It asks whether a *feature-provenance* choice can reverse a zero-shot verdict on identical flows.

CICIDS2017 ships a TrafficLabelling release that retains Protocol and Source Port, and a MachineLearningCVE release that omits both. Anyone aligning names to a nine-field InSDN schema must drop the absent fields or reconstruct them. We study one specified reconstruction as a controlled failure mode. The causal comparison is not InSDN versus CICIDS2017, and it is not the two public CICIDS2017 releases against each other: those releases differ in row set and prevalence. It is two encodings of the *same* 207,463 CICIDS2017 flows.

The central question is: can that reconstruction reverse a source-trained linear model's ranking on identical target rows, and which field drives the reversal? Source-domain FedAvg versus centralized and local training, and an IID versus Dirichlet non-IID partition, are reported as controls. They are not hypothesized as the cause; the non-IID model is not scored on the target views.

Our contributions are: (i) a row-matched counterfactual that changes only Protocol and Source Port, with the real MachineLearningCVE release as an unmatched ecological check and a centralized control showing the reversal is not an aggregation artifact; (ii) protocol-by-label evidence that the selected reconstruction collides source-domain encodings with opposite class associations; and (iii) masking, retraining, and a target-trained within-domain control that separate ranking reliance, missing-field policy, and remaining in-domain signal, together with a label-free screen that catches placeholder substitution but not this reconstruction.

II. Related Work and Research Gap

A. Federated intrusion detection and client heterogeneity

FedAvg established sample-weighted aggregation of local updates [1]. Non-IID clients cause objective drift, which FedProx, SCAFFOLD, and FedNova address [7]–[9]. FL-IDS work spans linear detectors [10], 5G studies with centralized and local controls [11], and deep or distillation-based defenses [12], [13]; dataset-centric evaluation finds in-domain scores near centralized ones while cross-dataset transfer degrades after label and feature alignment [14]. We keep plain FedAvg and logistic regression as an evaluation setting.

B. Cross-dataset NIDS and feature semantics

Dataset surveys show that collection, labeling, and flow extraction constrain valid comparisons [15]. Inter-dataset studies report large degradation even between related CIC corpora [3]; XeNIDS warns against assuming interchangeable flow fields [4]. Later work finds asymmetric transfer [5], poor cross-dataset performance despite strong in-domain scores [6], and dataset-dependent feature influence even under standardized NetFlow sets [17]. Closest to our setting, Demirbaş Paray and Aydos harmonize CICIDS2017 and InSDN/OVS on the ground that both came from the same extractor, so shared fields "carry identical semantic and technical definitions" [16]. Shared extractor lineage does not validate *every* field of a harmonized schema when a release omits columns and a reconstruction fills them.

C. Dataset defects, tooling, and release variants

CICIDS2017 has documented defects: flow-construction and labeling faults [18], packet misordering and an unlabeled port scan [19], and duplicated or miscalculated features with *wrong protocol detection* [20]. Heterogeneous-domain benchmarks materialize absent fields across *distinct* datasets [21]. Our gap is narrower: whether missingness induced by a public release of the *same* corpus, plus a specified reconstruction, can invert a cross-dataset conclusion while rows, labels, prevalence, and every unaffected field are held fixed.

III. Method

FedAvg aggregates local parameter vectors as $\mathbf{w}_{t+1} = \sum_{k=1}^{K} (n_k / \sum_j n_j)\, \mathbf{w}_{t+1}^k$ with $K = 5$ [1]. Each client runs one local pass of binary logistic regression per round. Server and clients start from a shared zero state. Centralized LR uses the same optimizer and 20 passes. Logistic regression is chosen so a sign reversal cannot be attributed to a deep architecture; it is not claimed to be an operational NIDS.

A. Feature-shift diagnostic

For source feature j we record the two-sample Kolmogorov–Smirnov distance $D_j = \sup_x |F_{sj}(x) - F_{tj}(x)|$ on the standardized column. KS compares empirical CDFs on the numeric order of values. Protocol codes $\{0, 6, 17\}$ are nominal labels, so that order is arbitrary; we therefore treat total variation (TV) between category proportions as the primary marginal comparison for protocol, and report KS only as a supplementary numeric-order statistic. Neither is a significance test: the study uses the distance, not a p-value.

We keep the trained IID global model fixed, replace one standardized feature by zero (the source-training mean), and compute $\Delta A_j = A_t^{\mathrm{mask}(j)} - A_t^{\mathrm{full}}$ on target ROC-AUC. Positive ΔA_j means masking *raises* ranking quality, so the learned contribution of that column harms target ranking; negative ΔA_j means the column carries transferable ranking signal. Masking is not retraining, not imputation, and not a causal effect of "removing" a raw field: correlated features share credit, and a constant column cannot reorder scores even if it moves a fixed-threshold metric.

B. Label-free provenance screen

From the source-training column x_j and the unlabelled target column $\tilde{x}_j$ the check flags a column when

$$\frac{\mathrm{sd}(\tilde{x}_j)}{\mathrm{sd}(x_j)} < \tau_v \quad \text{or} \quad \Pr[\tilde{x}_j \notin [\min x_j, \max x_j]] > \tau_r, \qquad (1)$$

with $\tau_v = \tau_r = 0.01$. A placeholder such as a constant sentinel typically triggers both. For integer-coded fields we also record TV between category proportions and a mass-collapse rule (a source category with mass $> 5\%$ falling below 1% on the target). These rules are compared with ΔA_j, not treated as causal ground truth.

TABLE I
THREE VIEWS OF THE SAME CICIDS2017 CAPTURE FILES. CF IS ROW-MATCHED TO TL; MLCVE IS THE REAL RELEASE AND IS NOT.

	TL (original)	CF (counterfactual)	MLCVE (real release)
Protocol field	original	flag surrogate	flag surrogate
Protocol values	$\{0, 6, 17\}$	$\{0, 6\}$	$\{0, 6\}$
Source port	original	sentinel -1	sentinel -1
Cleaned flows	1,037,315	1,037,315	880,176
Fixed test split	207,463	207,463	176,036
Sampled benign/attack	21,705 / 8,295	21,705 / 8,295	22,577 / 7,423
Attack prevalence	0.2765	0.2765	0.2474
Row-matched to TL	—	yes	no

IV. Experimental Protocol

A. Source data, mappings, and the three target views

Prepared InSDN [22] contains 343,889 records (68,424 normal; 275,465 anomaly). The stored protocol field takes integer codes $\{0, 6, 17\}$. We do *not* treat those integers as established IANA protocol identities: the files do not include packet-level confirmation, and code 0 in this InSDN extract is uniformly portless (source and destination port 0) with attack prevalence 99.7%. Codes 6 and 17 coincide with the usual TCP/UDP numbers in these CSVs, but that coincidence is not a proof of extractor semantics.

Each seed draws a stratified 30,000-record InSDN sample (24,000/6,000 train/test) and 30,000 records from each target view. Because TL and CF share row order and labels, one seeded draw selects the same rows from both.

All three target views describe the same CICIDS2017 capture files [23]: Monday benign, Friday DDoS, Friday PortScan. TL is built from TrafficLabelling and keeps the original Protocol and Source Port fields. MLCVE is built from MachineLearningCVE, which omits both; preserving a nine-field schema then requires a policy for the absent columns. We evaluate one lossy rule, chosen because TCP flags remain when Protocol does not, so a practitioner keeping the InSDN schema might infer "TCP versus other": protocol becomes 6 when any TCP flag is positive and 0 otherwise, and every source port becomes -1. It is a specified intervention, not a cited standard.

Comparing TL with MLCVE alone is confounded: the two releases survive cleaning and de-duplication differently (Table I). The CF view removes that confound. It is derived from the finished TL test split, overwriting only `ip_proto` and `tp_src`. Re-running the cleaner with those columns suppressed would change the row set, because missing-value filtering and de-duplication key on them. Scripts abort if the seven unaffected columns and labels are not identical or if more or fewer than two columns differ. The counterfactual identifies the effect of this two-field intervention on these rows and these linear models. It does not identify a property of federated averaging, of other imputations, or of NIDS transfer in general.

The nine model inputs—protocol, source and destination port, packet and byte counts, duration in seconds, packet and byte rates, and average packet size—are referenced by name, not position. InSDN supplies them from Protocol, Src/Dst Port, Tot

Fwd/Bwd Pkts, TotLen Fwd/Bwd Pkts, Flow Duration$/10^6$, Flow Pkts/s, Flow Byts/s, and Pkt Size Avg. CICIDS2017 supplies the corresponding TrafficLabelling names, or the flag surrogate and sentinel port when MachineLearningCVE omits those columns. Duration is treated as microseconds following CICFlowMeter documentation; rate and size fields are passed through. This is syntactic alignment of CICFlowMeter-family columns, not a proof that every formula is identical across extractor versions. A duplicated flow-duration column, identical to duration in seconds in every row [20], is stored but unused as a model input.

During preparation, infinities (and CICIDS string tokens Infinity/inf) become NaN, non-numeric entries are coerced to NaN, and rows with NaN in any used column or the label are dropped; missing forward or backward packet or byte counts are filled with 0 before that drop. Negative durations and rates are finite and are retained. At evaluation, any remaining non-finite values among the nine inputs are replaced by 0, then z-scored. A source-only standard scaler is fit on InSDN training features and applied unchanged to the InSDN test partition and all three target views. No target-domain fit, fine-tuning, or threshold selection is performed in the zero-shot evaluation. Labels: InSDN normal$\rightarrow$ 0, anomaly$\rightarrow$ 1 (sampled 5,969 benign / 24,031 attack); target normal$\rightarrow$ 0, ddos/portscan$\rightarrow$ 1.

B. Partitions and measures

IID FL splits shuffled training records equally among five clients; non-IID FL allocates each class by a Dirichlet distribution with $\alpha = 0.5$. Both run 20 rounds fixed *a priori*. Predicted labels use threshold 0.5; ROC-AUC and PR-AUC (trapezoidal area under the precision–recall curve) use scores. The random PR baseline is the attack prevalence, 0.801 in InSDN. At seed 11 the Dirichlet split yields client sizes 1,669 to 11,950 and benign shares 0.5% to 38%. Centralized LR uses the same 24,000 training records; local-only LR is an unweighted mean of five client models, not an ensemble. The classifier is scikit-learn SGD with log loss, $\eta = 0.01$, L2 $\alpha = 10^{-4}$, seeds 11, 23, 42, 77, 101, Python 3.11 / NumPy 2.2.6 / pandas 2.3.1 / scikit-learn 1.7.2.

V. Results

A. Source-domain controls

Table II reports mean$\pm$std over five repeated holdouts. FedAvg IID lies between centralized and local-only, with a gap smaller than seed-to-seed variation. Non-IID allocation preserves ROC-AUC while reducing balanced accuracy and MCC: an operating-point shift under class skew, not a ranking loss. Mean attack/benign recall is $0.9935/0.6635$ (centralized), $0.9941/0.6793$ (local-only), $0.9930/0.6781$ (IID) and $0.9995/0.6278$ (non-IID). These controls justify using this linear baseline on sampled InSDN; they do not show a federated accuracy advantage, and they do not explain the target-view reversal below.

TABLE II
SOURCE-DOMAIN (INSDN) FIVE-SEED RESULTS; MEAN$\pm$STD.

Setting	F1	ROC-AUC	Bal. acc.	MCC
Central	0.9567±0.0033	0.9320±0.0038	0.8285±0.0149	0.7624±0.0199
Local-only	0.9588±0.0027	0.9288±0.0030	0.8367±0.0120	0.7751±0.0162
FedAvg IID	0.9580±0.0023	0.9324±0.0039	0.8355±0.0091	0.7709±0.0140
FedAvg non-IID	0.9556±0.0038	0.9336±0.0064	0.8136±0.0166	0.7562±0.0228

TABLE III
ZERO-SHOT CROSS-DATASET RESULTS, FIVE SEEDS, NO REFITTING. CF DIFFERS FROM TL IN EXACTLY TWO COLUMNS ON IDENTICAL ROWS; BOLD MARKS THAT REVERSAL. CHANCE IS 0.5 FOR ROC-AUC AND BALANCED ACCURACY, 0 FOR MCC, AND TABLE I PREVALENCE FOR PR-AUC.

Target view	ROC-AUC	PR-AUC	Bal. acc.	MCC
FedAvg IID global model				
TL (original)	**0.7357±0.0070**	0.4520±0.0154	0.6403±0.0037	**0.2511±0.0065**
CF (matched)	**0.3545±0.0031**	0.2102±0.0009	0.4191±0.0037	**−0.1915±0.0097**
MLCVE (real)	0.3441±0.0040	0.1848±0.0010	0.3766±0.0023	−0.2698±0.0067
Centralized model, same records				
TL (original)	**0.7490±0.0165**	0.4656±0.0225	0.6336±0.0047	**0.2392±0.0083**
CF (matched)	**0.3544±0.0033**	0.2100±0.0010	0.4163±0.0045	**−0.2004±0.0147**
MLCVE (real)	0.3447±0.0048	0.1848±0.0012	0.3738±0.0074	−0.2782±0.0198

B. Reconstruction reverses the verdict on identical rows

The seven-field intersection of Section V-D is provenance-valid, not predictively adequate. The nine-field comparison isolates the cost of this reconstruction. The same five IID FedAvg models are scored on all three views (Table III). On TL, transfer is degraded ($0.9324 \rightarrow 0.7357$ ROC-AUC) but remains informative (MCC $+0.2511$, PR-AUC 0.4520 against prevalence 0.2765). Overwriting only the two encodings sends ROC-AUC, PR-AUC, balanced accuracy, and MCC below their no-skill baselines, and F1 to 0.3558.

The real MLCVE release lands close to the counterfactual (0.3441 against 0.3545) but differs in row set and prevalence, so it corroborates rather than identifying the matched effect. The centralized model shows the same reversal ($0.7490 \rightarrow 0.3544 \rightarrow 0.3447$).

C. Protocol codes and label associations

Table IV reports stored integer codes against binary labels on the prepared files. In the TL test split, all 62,901 code-17 (UDP-coded) rows are benign, and every DDoS row has code 6. PortScan is not exclusive to code 6: three PortScan rows have code 0, which is the 3.7% attack rate on TL code 0 in Table IV; those three rows remain after reconstruction, so CF code 0 is not exactly attack-free. InSDN associates code 0 with 99.7% attacks and code 17 with 0.3% attacks. Every protocol disagreement between TL and CF is $17 \rightarrow 0$ (30.32% of rows). The surrogate therefore writes the source's strongly attack-associated code onto the target slice's entirely benign code-17 population. That is a semantic collision in the source-model feature space, not a change of the underlying flows. We do not claim the fitted coefficients implement a perfect decision rule from protocol alone; masking and retraining below test reliance.

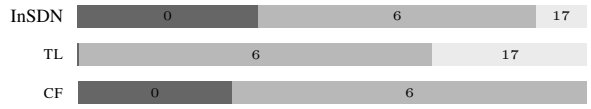


Fig. 1. Share of stored protocol codes 0 (dark), 6, and 17 (light) in prepared InSDN and the row-matched 207,463-row TL/CF split. Reconstruction removes code 17.

TABLE IV
STORED PROTOCOL CODES VERSUS ATTACK RATE ON PREPARED FILES. INSDN $n = 343{,}889$; TL/CF $n = 207{,}463$. CODE 0 IS NOT IDENTIFIED AS A NETWORK PROTOCOL.

Code	InSDN share (att.)	TL share (att.)	CF share (att.)
0	35.4% (99.7%)	0.04% (3.7%)	30.4% (<0.1%)
6	54.7% (81.9%)	69.6% (39.7%)	69.6% (39.7%)
17	9.9% (0.3%)	30.3% (0%)	—

D. Localizing the reversal

Masking the fixed model localizes the ranking reversal to protocol, the only field whose ΔA changes sign: on TL, masking protocol costs 0.1438 ROC-AUC ($\Delta A = -0.1438$), whereas on CF masking the surrogate *gains* 0.2647. The real release agrees ($+0.2103$). Protocol TV is 0.3532 on TL and 0.1489 on CF—the harmful encoding looks *closer* to InSDN on identical rows. Numeric-order KS repeats the direction (0.3532 versus 0.0992) but is not the appropriate categorical comparison.

Source port is a different failure. Masking it *gains* 0.0165 ROC-AUC on TL ($\Delta A = +0.0165$), so its learned contribution harms TL ranking slightly; we include the field because it is part of the InSDN schema and is the other MLCVE-absent column, not because it is a justified NIDS predictor. On CF every value is the sentinel -1, KS is maximal, and masking leaves ROC-AUC unchanged to four decimals: a constant adds the same amount to every logit and cannot reorder scores. It does move the operating point, shifting CF MCC by $+0.0186$. Protocol therefore localizes the ranking reversal; source port does not.

Retraining is a different intervention (Table V). Dropping protocol before training costs source F1 ($0.9580 \rightarrow 0.9118$) and collapses the TL–CF gap (0.4953, 0.5157, 0.4668). On matched CF rows, dropping the unavailable field reaches 0.5157 while reconstructing it reaches 0.3545. One-hot encoding protocol is the best TL configuration (0.7472) yet barely moves CF (0.3630). Removing both ports lowers TL ROC-AUC to 0.7062; masking attributes most of that ranking loss to destination port ($\Delta A = -0.0605$), which every view supplies. The seven-field intersection drops both MLCVE-absent fields rather than reconstructing them: source F1 0.9082, chance-level transfer 0.5075, and TL=CF to every reported digit. In this experiment, dropping unverified unavailable fields and retraining is a conservative provenance-valid baseline. It is not a useful detector, and it is not shown to dominate every alternative imputation.

TABLE V
RETRAINING ABLATIONS AND A TARGET-TRAINED WITHIN-DOMAIN LR CONTROL, FIVE SEEDS, ROC-AUC. SOURCE COLUMNS ARE INSDN. CHANCE IS 0.5. PANEL (B) SEED DISPERSION IS LARGER (STANDARD DEVIATIONS 0.055–0.093).

Feature set	Src. F1	Src. AUC	TL	CF	MLCVE
(a) Zero-shot transfer: source model trained per feature set, no target refitting					
All nine (reconstructed)	0.9580	0.9324	0.7357	**0.3545**	**0.3441**
one-hot protocol	0.9582	0.9311	0.7472	0.3630	0.3506
without ports	0.9583	0.9267	0.7062	0.3005	0.2857
without protocol	0.9118	0.8915	0.4953	0.5157	0.4668
Intersection, 7 fields	0.9082	0.8954	0.5075	0.5075	0.4505
(b) Target-trained within-domain control					
All nine	—	—	0.8995	0.8820	0.8956
Intersection, 7 fields	—	—	0.7200	0.7200	0.7117

E. Target-trained within-domain control

Zero-shot scores cannot separate lost information from changed meaning. Table V(b) trains and evaluates logistic regression inside one view (its own scaler, 80/20 of that view's 30,000-row sample). This is a within-domain diagnostic for this model class, not an information-theoretic upper bound.

On the matched TL/CF samples the control reaches 0.8820 ± 0.0630 against 0.8995 ± 0.0553, while zero-shot transfer is 0.3545 against 0.7357. Pairing by seed, the in-domain gap is 0.0175 (95% CI $[0.0069, 0.0280]$, TL ahead on all five seeds) and the zero-shot gap is 0.3813 ± 0.0052 on the 30,000-row samples. Reconstruction therefore costs a small, consistent amount of in-domain LR signal and changes much more of what the source encoding means. The real MLCVE release agrees (0.8956 in-domain, 0.3441 transfer). The seven-field control reaches 0.7200 in-domain on TL and CF and 0.7117 on MLCVE, which the source model does not capture at 0.5075.

F. What a label-free preflight check can and cannot catch

Among 27 column–view pairs the screen flags two on all five seeds and no others: source port of CF and of MLCVE (ratio 0.0000, out-of-range 1.0000). No unflagged column falls below ratio 0.352 or exceeds out-of-range 0.0028; the byte-rate column is 31 to 48 times more dispersed yet has ΔAUC 0.0000.

The reconstructed protocol keeps a plausible dispersion ratio (0.5672 on CF) and never leaves the source range. TV scores the harmful encoding as closer (0.1489 against 0.3532). The mass-collapse rule fires on all three views, including uncorrupted TL, where protocol 0 falls from 35% of source mass to 0.04%. Silence is not evidence of sound provenance; metadata recorded at preparation time is.

VI. DISCUSSION

The matched contrast cannot be attributed to attack mix, labeling, collection environment, sampling, or FedAvg averaging: rows, labels, prevalence, seven features, scaler, and threshold are shared, and the centralized model reverses too. The reversal is consistent with a protocol-dependent shortcut in this linear model on this slice: the reconstruction maps the target's benign code-17 population onto InSDN's attack-associated code 0. Rosay et al. report wrong protocol detection

in these releases [20]; we quantify what one reconstruction of that field does to a verdict.

A provenance-valid intersection is not a useful detector: 0.5075 is an honest non-answer. Other imputations (mode, missing indicator, unknown category) are untested. Non-IID source results in Table II show that heterogeneity can move operating-point metrics without moving ROC-AUC; they do not make reconstruction an FL mechanism.

VII. Threats to Validity and Reproducibility

The study uses tabular flows, not packets, one source corpus, one restricted CICIDS2017 slice, one linear model, five repeated holdouts from a finite corpus, and one reconstruction rule. Whether the reversal appears for trees, neural models, other remedies, or other dataset pairs is open. Protocol code 0 is characterized empirically, not identified at the packet layer. Masking reports model reliance, not a unique causal feature effect. Binary labels and the 0.5 threshold limit external validity. The non-IID model is not evaluated on the target views.

VIII. Artifact Availability

Preparation records per-file read-or-reconstructed status and publishes SHA-256 checksums. Evaluation regenerates five-seed outputs under the pinned environment and aborts if CF is not row-matched. A verifier checks quoted numbers against committed outputs. The artifact is at https://github.com/thtcsec/rivf2026-federated-edge; third-party datasets are not redistributed.

IX. Conclusion

On one fixed set of 207,463 CICIDS2017 flows, a specified reconstruction of two MachineLearningCVE-absent fields moves InSDN-trained logistic regression from ROC-AUC 0.7357 to 0.3545. A centralized model reproduces the reversal, so federated averaging is not the cause. The selected slice labels every code-17 (UDP-coded) flow benign, while InSDN associates code 0 with 99.7% attacks; the surrogate collides those encodings. Dropping the unavailable fields avoids the below-random result but transfers at chance (0.5075). Cross-dataset NIDS evaluations should report feature provenance per field.

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